Weighted sum Model

Which car to buy?

Attributes	Price (In lakhs)	Fuel Efficiency(Km/Lit	Comfort	Looks
Car1	4	15	Good	Average
Car2	8	16	Good	Good
Car3	10	17	Excellent	Excellent
Car4	3.5	16	Average	Average

Weighted sum Model (WSM) or Simple Additive Weighting (SAW)

Lets take an example in which DM has to select a car out of 4 possible options in the market.

Attributes →	**Price (In lakhs)**	**Fuel Efficiency(Km/ Lit**	**Comfort**	**Looks**
Car1	4	15	Good	Average
Car2	8	16	Good	Good
Car3	10	17	Excellent	Excellent
Car4	3.5	16	Average	Average

Step 1

Convert the linguistic terms in a scale of 1-5 or 1-10.

- In this example converting comfort and looks in a scale of 1-5.
- Average -1, Good-2, Very Good-3, Excellent-4, Oustanding-5

Attributes	**Price (In lakhs)**	**Fuel Efficiency (Km/Lit**	**Comfort**	**Looks**
Car1	4	15	Good	Average
Car2	8	16	Good	Good
Car3	10	17	Excellent	Excellent
Car4	3.5	16	Average	Average

Attributes	**Price (In lakhs)**	**Fuel Efficiency(Km/Lit**	**Comfort**	**Looks**
Car1	4	15	2	1
Car2	8	16	2	2
Car3	10	17	4	4
Car4	3.5	16	1	1

Step 2: Categorization of attributes-

- ✓Benefit attributes- More the value more preferred the attribute .
- ✓Cost attributes- Greater the attribute value the less its preference.

Attributes	Price (In lakhs)	Fuel Efficiency(Km/Lit	Comfort	Looks
	COST	**BENEFIT**	**BENEFIT**	**BENIFIT**
Car1	4	15	Good	Average
Car2	8	16	Good	Good
Car3	10	17	Excellent	Excellent
Car4	3.5	16	Average	Average

Step 3: Normalization- Use any appropriate normalization method

(Here we are using "Linear-scale transformation (Sum)" method)

for benefit attributes=x(ij)/Sum(xij)

for cost attributes=1/x(ij)/Sum(1/(xij))

Attributes	**Price**	**nv**	**Fuel Efficiency**	**nv**	**Comfort**	**nv**	**Looks**	**nv**
Car1	.25	0.33	15	0.23	2	0.22	1	.13
Car2	.125	0.16	16	0.25	2	0.22	2	.25
Car3	0.1	0.13	17	0.27	4	.44	4	.5
Car4	0.29	0.38	16	0.25	1	.11	1	.13
Sum	Sum (1/x(ij)) =0.765		Sum (x(ij))=64		Sum (x(ij))=9		Sum (x(ij)) =8	

After Step 3:

Normalized decision Matrix

Attributes	Price	Fuel Efficiency	Comfort	Looks
Car1	0.33	0.23	0.22	.13
Car2	0.16	0.25	0.22	.25
Car3	0.13	0.27	.44	.5
Car4	0.38	0.25	.11	.13

Step 4 : Weight calculation of attributes

Use any appropriate method for calculation of weights(if not given).

Using AHP-

	Price	Fuel Efficiency	Comfort	Looks
Weights	0.604	0.136	0.196	0.064

Step 5

- The global score (preference score) is calculated for each alternative using

$$v(A_i) = \sum_{j=1}^{n} w_j r_{ij}$$

Attributes	Price	Fuel Efficiency	Comfort	Looks
Weights	0.604	0.136	0.196	0.064
Car1	0.33	0.23	0.22	.13
Car2	0.16	0.25	0.22	.25
Car3	0.13	0.27	.44	.5
Car4	0.38	0.25	.11	.13

Attributes	Price	Fuel Efficiency	Comfort	Looks	$v(A_i)$
Weights	0.604	0.136	0.196	0.064	
Car1	0.33*0.604	0.23*0.136	0.22*0.196	.13*0.064	0.28204
Car2	0.16*0.604	0.25*0.136	0.22*0.196	.25*0.064	0.18976
Car3	0.13*0.604	0.27*0.136	.44*0.196	.5*0.064	0.23348
Car4	0.38*0.604	0.25*0.136	.11*0.196	.13*0.064	0.2934

Step 5

- The global score (preference score) is calculated for each alternative using

$$v(A_i) = \sum_{j=1}^{n} w_j r_{ij}$$

Attributes	Price	Fuel Efficiency	Comfort	Looks
Weights	0.604	0.136	0.196	0.064
Car1	0.33	0.23	0.22	.13
Car2	0.16	0.25	0.22	.25
Car3	0.13	0.27	.44	.5
Car4	0.38	0.25	.11	.13

Attributes	Price	Fuel Efficiency	Comfort	Looks	$v(A_i)$
Weights	0.604	0.136	0.196	0.064	
Car1	0.33*0.604	0.23*0.136	0.22*0.196	.13*0.064	0.28204
Car2	0.16*0.604	0.25*0.136	0.22*0.196	.25*0.064	0.18976
Car3	0.13*0.604	0.27*0.136	.44*0.196	.5*0.064	0.23348
Car4	0.38*0.604	0.25*0.136	.11*0.196	.13*0.064	0.2934

Step 6

Select maximum $v(A_i)$

Attributes	Price	Fuel Efficiency	Comfort	Looks	$v(A_i)$	Preference Rank
Weights						
Car1					0.28204	2
Car2					0.18976	4
Car3					0.23348	3
Car4					0.2934	1

Selected car using WSM